CNS ALA-3 • INFOGRAPHIC

SOCIAL ENGINEERING

ATTACKS & MAJOR SECURITY BREACHES



Real incidents • Attack techniques • Prevention strategies

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Three ways trust can be exploited

These methods manipulate people into revealing information, granting access, or taking unsafe actions.

01

PHISHING

Fake message or website

An attacker imitates a trusted organisation and urges the target to click a link, open an attachment, or share login details.

CHECK: sender • URL • urgency

02

PRETEXTING

A made-up believable story

An attacker pretends to be IT support, a bank employee, or a colleague to persuade someone to disclose information or approve access.

CHECK: verify identity independently

03

BAITING

A tempting offer or item

A fake reward, download, or unknown USB device encourages someone to install unsafe content or expose information.

CHECK: avoid unknown devices / offers

Equifax data breach • USA, 2017

A major example of how an unpatched vulnerability can expose sensitive personal data.

SCALE OF IMPACT

~147 MILLION

people affected (later estimate)

Sensitive data exposed included names, addresses, dates of birth, Social Security numbers and some payment-card information.

WHAT HAPPENED

Attackers exploited a known vulnerability in an internet-facing Apache Struts application and remained in the environment for weeks before discovery.

RESPONSE

Investigation, consumer notification, identity-theft / credit-monitoring assistance, and stronger vulnerability-management controls.

HOW TO PREVENT IT

Patch critical vulnerabilities quickly • scan internet-facing systems • segment sensitive databases • monitor unusual data access.

FINANCIAL NOTE: FTC settlement required at least US$575 million, potentially up to US$700 million.

Optus and Tribune Publishing

Different impacts: exposure of customer data versus disruption of essential operations.

AUSTRALIA • 2022

OPTUS DATA BREACH

WHAT HAPPENED

A cyberattack affected current and former customers. Information such as names, dates of birth, contact details and some identity-document numbers was exposed.

PREVENTION

Minimise stored personal data, secure APIs, apply least privilege, test systems, and monitor unusual data extraction.

Reported response expense: about A$140 million.

USA • 2018

TRIBUNE / LA TIMES MALWARE

WHAT HAPPENED

Malware spread through a shared production environment, disrupting newspaper printing and delaying editions. The initial entry method was not publicly confirmed.

PREVENTION

Network segmentation, endpoint detection, regular patching, tested offline backups and rehearsed business-continuity plans.

Main impact: operational disruption.

Fraud, stolen card data & human trust

The supplied reference describes broad cybercrime patterns; not every case has a confirmed single entry technique.

INDIA • CYBER-FRAUD

DELHI CYBER-FRAUD SURGE

TECHNIQUES

Investment scams, fake work-from-home offers, digital-arrest threats and other pretexting methods build trust or urgency to trick people into sending money.

PREVENTION

Reject guaranteed returns, independently verify platforms, never share OTP/PIN/passwords, and report financial fraud immediately via 1930.

INDIA • CARD DATA

CARD DATA ON DARK-WEB MARKETS

WHAT WAS REPORTED

A 2019 article reported information relating to about 1.3 million Indian debit and credit cards being offered for sale. The exact loss tied to that dataset was not known.

PREVENTION

Use transaction alerts and strong authentication; never share OTP/PIN/CVV; replace compromised cards; protect databases with encryption and strict access controls.

Break the chain before damage spreads

A simplified phishing flow — and the control that can stop each stage.

1 • CONTACT
Fake email / message

2 • TRUST
Target believes it

3 • ACTION
Clicks / shares details

4 • IMPACT
Account / data exposed

BREAK THE CHAIN: Pause → verify independently → report → reset exposed credentials

AWARENESS
Recognise suspicious requests and urgency.

PATCHING
Fix critical vulnerabilities quickly.

LEAST PRIVILEGE
Limit user, vendor and admin access.

MONITORING
Alert on unusual logins or data transfers.

BACKUPS
Keep offline backups and test restoration.

FAST REPORTING
In India, report financial fraud via 1930.